

ECON 311

Lecture 7: Solow Growth Model, part II

Capital Accumulation

- Investing in capital for $t + 1$ requires resources at t :

$$C_t + I_t = Y_t \quad (\text{resource constraint})$$

- ▶ An assumption of the Solow model is that capital and consumption goods are produced using the same production function.
- We are going to assume that individual decisions to save and invest is exogenous:

$$I_t = \bar{s}Y_t \quad (\text{allocation of resources}) \quad , \quad 0 \leq \bar{s} \leq 1$$

- Combining the allocation of resources with the capital accumulation equation, we obtain:

$$\Delta K_{t+1} = \bar{s}Y_t - \bar{d}K_t \quad (\text{law of motion})$$

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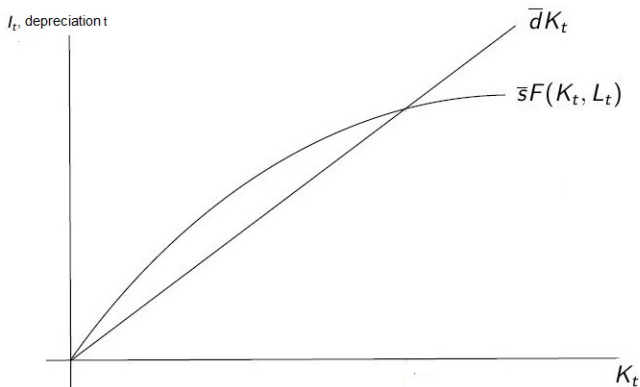
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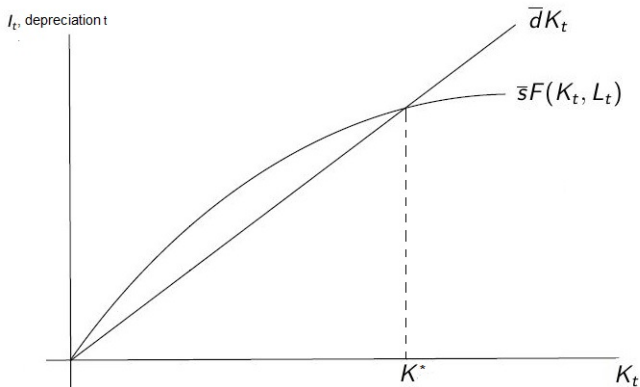
$$\Delta K_{t+1} = \bar{s}F(K_t, L_t) - \bar{d}K_t \quad (\text{law of motion})$$

Dynamics of Capital Accumulation



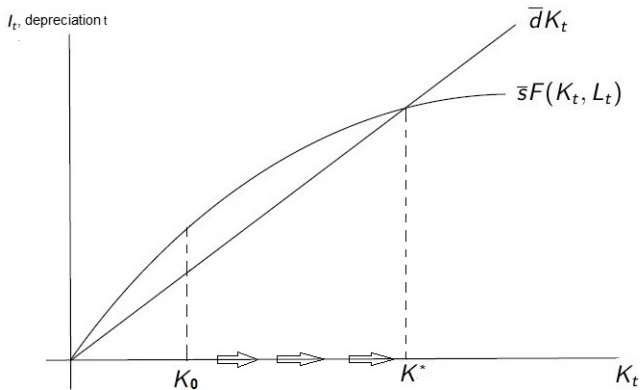
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Dynamics of Capital Accumulation



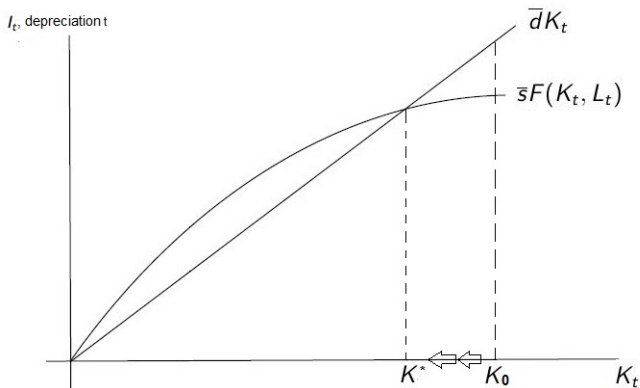
$$\underbrace{\Delta K_{t+1}}_{K_{t+1}=K_t} = 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} = \bar{d}K_t$$

Dynamics of Capital Accumulation



$$\underbrace{\Delta K_{t+1}}_{K_{t+1} > K_t} > 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} > \bar{d}K_t$$

Dynamics of Capital Accumulation



$$\underbrace{\Delta K_{t+1}}_{K_{t+1} < K_t} < 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} < \bar{d}K_t$$

Dynamics of Capital Accumulation

- To simplify the analysis, we are going to assume that: $L_t = \bar{L}$
- At K^* , we say that the economy is at its steady-state equilibrium.
- We can solve for K^* as a function of exogenous variables and parameters:

$$K^* = \bar{L} \left(\frac{\bar{s}\bar{A}}{\bar{d}} \right)^{\frac{3}{2}}$$

or in per capita terms: $k^* = \left(\frac{\bar{s}\bar{A}}{\bar{d}} \right)^{\frac{3}{2}}$

- We are also able to solve for the steady-state level of output:

$$Y^* = \bar{L} \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{\bar{d}} \right)^{\frac{1}{2}}$$

or in per capita terms: $y^* = \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{\bar{d}} \right)^{\frac{1}{2}}$

Cross-country differences in $\bar{s}$ and growth

- Suppose there are two countries, CAN and ARG, with the same $\bar{A}$, $\bar{L}$, $\bar{d}$ and K_0 but with different $\bar{s}$

$$\bar{s}_{CAN} > \bar{s}_{ARG}$$

- At $t = 0$ both countries will have the same level of output per worker:

$$y_0 = \bar{A} \left(\frac{K_0}{\bar{L}} \right)^{\frac{1}{3}}$$

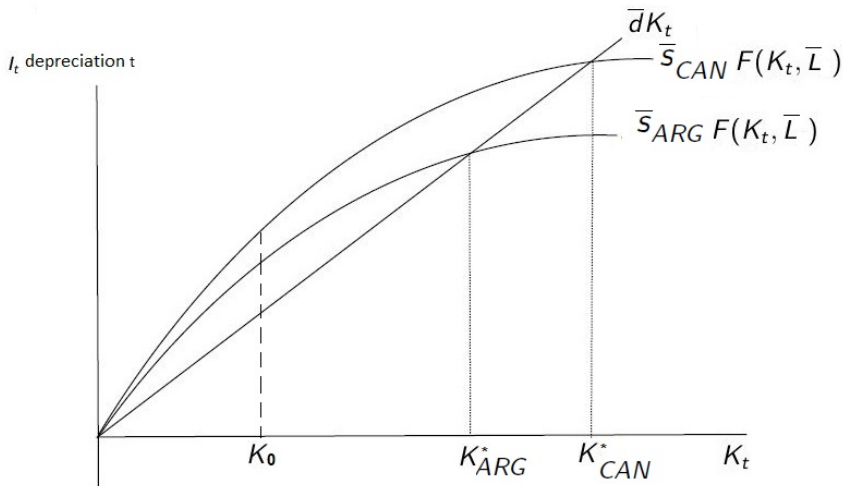
- However, in the long run:

$$y_{CAN}^* = \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}_{CAN}}{\bar{d}} \right)^{\frac{1}{2}} > \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}_{ARG}}{\bar{d}} \right)^{\frac{1}{2}} = y_{ARG}^*$$

Cross-country differences in $\bar{s}$ and growth

- Since the starting level of GDP per capita is the same but the level of GDP per capita is higher in Canada in steady state:
 - ▶ GDP per capita on average over the long run grows faster in Canada than in Argentina.
- The main reason for this difference in growth rates is that capital accumulation is higher in Canada.

Cross-country differences in $\bar{s}$ and growth



Solving for Equilibrium

- For any t , the aggregate economy in the Solow Model is characterized by five equations and five unknowns (K_t, L_t, I_t, C_t, Y_t):

$$\Delta K_{t+1} = \bar{s}F(K_t, L_t) - \bar{d}K_t \quad (\text{law of motion})$$

$$Y_t = \bar{A}K_t^{\frac{1}{3}}L_t^{\frac{2}{3}} \quad (\text{production function})$$

$$C_t + I_t = Y_t \quad (\text{resource constraint})$$

$$L_t = \bar{L} \quad (\text{labour force})$$

$$I_t = \bar{s}Y_t \quad (\text{allocation of resources})$$

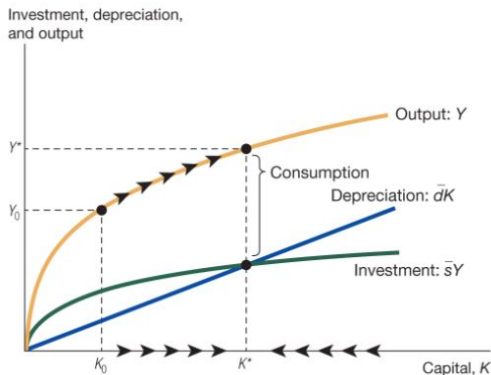
- For an initial level of capital, K_0 , we can solve for the endogenous variables in any time period.
 - Instead of solving the model for any time period, which requires a lot of algebra, we will only solve analytically for the steady state (ss) and provide graphical interpretation for other time periods.

Output and Consumption per capita

- An important variable in the Solow Model is the level of consumption at time t , which can be written as:

$$C_t = (1 - \bar{s})Y_t$$

The Solow Diagram with Output

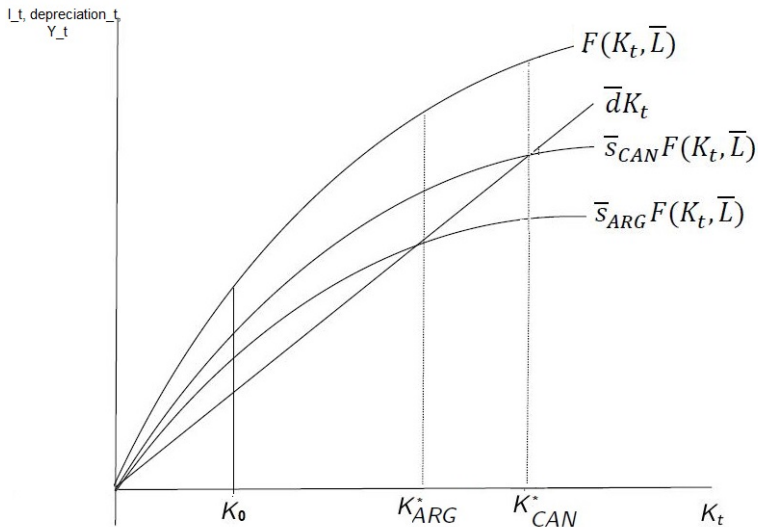


Output and Consumption per capita

- The resource constraint formalizes the trade-off an economy faces:
 - ▶ In order to grow faster and achieve higher production and higher consumption in the long run, initial consumption must be lower.
- We can see this point in our example of the Canadian and Argentine economies:

$$t = 0 \quad c_0^{ARG} = (1 - \bar{s}_{ARG})\bar{A} \left(\frac{K_0}{L} \right)^{\frac{1}{3}} > (1 - \bar{s}_{CAN})\bar{A} \left(\frac{K_0}{L} \right)^{\frac{1}{3}} = c_0^{CAN}$$

Output and Consumption per capita



Solving for Steady State

- To solve for the steady state equilibrium, we can reduce the equilibrium conditions to three equations and three unknowns:

$$\Delta K_{t+1} = 0 \quad \Rightarrow \quad K^* = \bar{L} \left(\frac{\bar{s}\bar{A}}{d} \right)^{\frac{3}{2}}, \quad k^* = \left(\frac{\bar{s}\bar{A}}{d} \right)^{\frac{3}{2}}$$

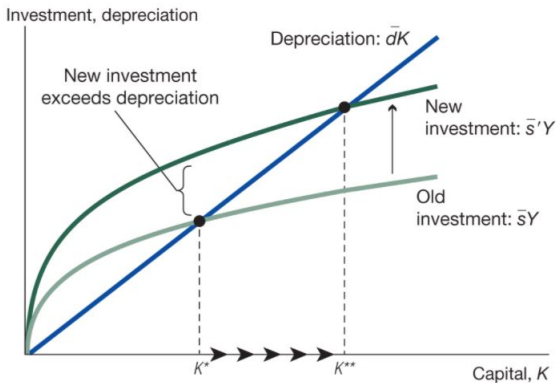
$$Y^* = \bar{A}(K^*)^{\frac{1}{3}} (\bar{L})^{\frac{2}{3}} \quad \Rightarrow \quad Y^* = \bar{L} \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}, \quad y^* = \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}$$

$$C^* = (1 - \bar{s})Y^* \quad \Rightarrow \quad C^* = (1 - \bar{s})\bar{L} \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}, \quad c^* = (1 - \bar{s})\bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}$$

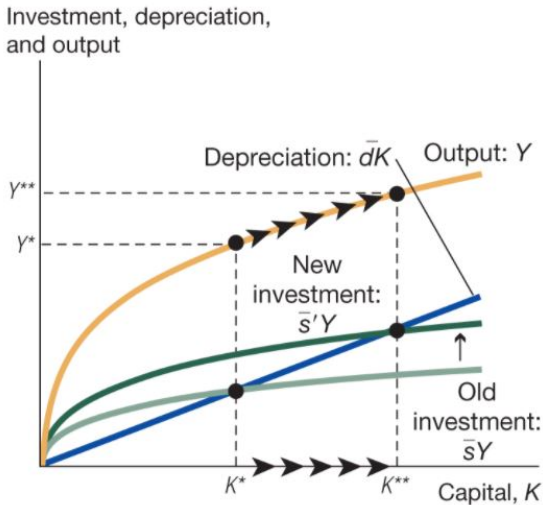
- We will also study how the system behaves out of the steady state without solving for endogenous variables.
 - ▶ This is important to understand differences in growth rates.

An increase in savings rate $\bar{s}$

- Assume that the economy is in its steady-state equilibrium and in 2020, there is an increase in savings:



An increase in savings rate $\bar{s}$



Transition Dynamics

